

Experimental and numerical study of strain distributions and shear lag effects in FRP tendon pull-out experiments

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ABSTRACT

In the present study, sand-coated carbon fiber reinforced polymer (CFRP) tendons were instrumented with integrated optical fibers. The direct integration during manufacturing did not affect the bond between tendon and concrete and resulted in an excellent measurement quality. The tendons were cast into concrete cubes with different embedment lengths of 5, 10 and 15 times the diameter. The bond between tendon and concrete was tested using direct pull-out tests. Not only the force and the end-slip were measured but also the strain distribution in the tendon. This allowed to derive a position-dependent local bond-slip curve. Significant deviations from the commonly measured average bond-stress vs. end-slip relationship were observed for long and short embedment lengths. A complementary finite element study confirmed that average bond-stress vs. end-slip curves cannot properly predict the strain distribution in pull-out experiments, especially for longer embedment lengths. This allows to conclude that strain distributions are highly affected by local effects, e.g. partial pull-out failure of the concrete. In addition, shear lag was identified as additional parameter affecting the measurements of optical fibers when integrated in materials with low shear stiffness. The shear lag is highly dependent on the radial and axial position and further affected by the tendon material, the diameter of the tendon and the magnitude of the shear stress.

1. Introduction

The load transfer between reinforcement and concrete is one essential aspect in structural engineering and has attracted a large amount of research over the last century. The bond-stress vs. slip behavior (τ - δ) of fiber reinforced polymer (FRP) rebars was measured using different standardized or non-standardized procedures [1–6]. Commonly, an average bond stress (τ_{avg}) over the embedment length is assumed, which results in a linear normal strain distribution of the tendon. The shorter the embedment length is, the better the approximation by a constant bond stress [7]. Therefore, standards define the embedment length usually to 5 times the rod or tendon diameter [8,9]. Nonetheless, the significance of a τ - δ relationship defined by evaluation of a direct pull-out tests is controversial. Sand-coated FRP in particular show very stiff bond behavior with high maximum bond stresses. Transfer lengths calculated with average bond-stresses τ_{avg} can be significantly shorter than true transfer lengths in prestressed elements. The study of the

local τ - δ relationship during pull-out experiments is therefore of importance. This will allow a deeper understanding of the bond mechanisms. However, this requires high resolution measurement of the axial strain distribution. In the past, strain gauges were the first choice for analyzing the load transfer between reinforcement and concrete. Abrishami and Mitchell used strain gauges applied on the surface of steel reinforcement bars to characterize the bond stress along the embedment length. The ability of controlling the force on both sides of the specimen allowed them to characterize the bond also in the softening branch, overcoming the problems of potential sudden pull-out failures [10]. Viathanatepa et al. studied the effects of generalized loadings on the bond of reinforcing bars embedded in concrete blocks. The instrumentation with several strain gauges allowed them to estimate the local bond stress [11]. Nanni et al. applied strain gauges on an aluminium tube, which was then integrated into a glass fiber reinforced polymer (GFRP) rod assuming a perfect bond. This allowed them to measure

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Nomenclature**Abbreviations**

CFRP	Carbon fiber reinforced polymer
DFOS	Distributed fiber optic sensing
FBG	Fiber Bragg gratings
FE	Finite element
FRP	Fiber reinforced polymer
GFRP	Glass fiber reinforced polymer
HPC	High performance concrete
LLS	Linear least square
PMMA	Polymethylmethacrylate

Greek letters

δ	Slip
δ_0	Linear limit of traction–separation response
δ_i	Separation, i-direction
δ_{end}	End slip
ϵ_c	Strain of concrete
ϵ_t	Strain of tendon
ϵ_{ii}	Normal strain, i-direction
ν	Poisson's ratio
ν_{fij}	Poisson's ratio of fiber, ij-component
ν_{ij}	Poisson's ratio of composite, ij-component
ν_m	Poisson's ratio of matrix
ρ_t	Reinforcement content
$\Delta\sigma_{11}$	Shear lag normal stress
σ_{fult}	Tensile strength of fibers
σ_{ii}	Normal stress, i-direction
τ	Bond stress
τ_{avg}	Average bond stress
τ_{ij}	Shear stress, ij-direction

Latin letters

C_{ii}	Compliance, i-direction
D	Damage variable
d	Nominal diameter of tendon
E	Young's modulus
E_c	Young's modulus of concrete
E_m	Young's modulus of matrix
E_t	Young's modulus of tendon
E_{fii}	Young's modulus of fiber, i-direction
E_{ii}	Young's modulus of composite, i-direction
F_{ii}	Force, i-direction
G_{fij}	Shear modulus of fiber, ij-direction
G_{ij}	Shear modulus of composite, ij-direction
K_i	Initial stiffness of τ - δ -curve
K_{ij}	Elasticity component, ij-direction
L_b	Embedment length
m	Constant
n	Constant
q_x	Line distributed force
r	Radius
$\bar{t}_i$	Traction stress component without damage, i-direction
t_i	Traction stress component, i-direction
t_i^0	Peak values of the nominal stresses, i-direction

u_c	Displacement of concrete
u_t	Displacement of tendon
V_f	Fiber volume fraction
x	Location
$\tilde{x}$	Corrected location

the strain at discrete locations without any local disturbances of the stress transfer [12]. This method has limited resolution due to space restrictions and is only available for larger tendon diameters, which enable the production of a central hole with sufficient diameter. Furthermore, the instrumentation changes the stiffness of the tendon in axial and radial direction. Lemcherreq et al. gave an overview about the instrumentation of steel reinforcement using strain gauges, fiber Bragg gratings (FBG) and distributed fiber optic sensing (DFOS) [13]. The optical fibers are typically glued into grooved rebars. The groove and the adhesive protects them from the concrete and mechanical damage. In [14], DFOS were successfully applied for the measurement of strain distributions in pull-out tests under monotonic and repeated loading for steel rebars. Rolland et al. performed direct pull-out experiments of FRP using DFOS integrated in a groove [15]. Measurements based on DFOS bonded to the surface show a relatively high noise level, as reported by [14–16] and require significant post-processing of the measurement data. Furthermore, the bond stress is negatively influenced by the groove, which is necessary for the integration of the fiber [17]. This effect is thought to be more pronounced for smaller rod diameters, as the dimensions of the groove are typically in the range of $0.8 \times 0.8 \text{ mm}^2$, independently of the rod diameter. One possibility to overcome all these limitations is the integration of optical fibers in the FRP tendon during manufacturing. Nellen et al. produced instrumented tendons for CFRP parallel wire bundles of cable-stayed bridges in a pultrusion process [18]. One further advancement was given by the development of a process for the subsequent extraction of the optical fibers from a cured FRP tendon [19]. This process allows to prepare also short specimens with a reasonable expense of time and effort. Nonetheless, the handling of the optical fiber with the missing protecting coating, which gets decomposed during the heating cycle, is crucial. Apart from the advantages of optical fibers, they also have their pitfalls. For both, externally bonded or directly integrated sensors, a perfect bond is assumed and can only be guaranteed by a proper preparation of the substrate surface and the use of an appropriate adhesive, which is typically epoxy-based. Nonetheless, the measurement is affected by the adhesive which has finite dimensions. The stiffness and the thickness of the adhesive can distort the measurement [20]. Apart from the adhesive, the coating of the optical fiber which is an inevitable component affects the measurement. The coating reduces the risk of failure of the susceptible optical fiber during handling. While for polyimide coated fibers a chemical bonding guarantees an excellent bond, acrylate based coatings rely on a purely mechanical bond and show significant slip. Therefore, not only the stiffness of the coating but especially its bond to the optical glass fiber can lead to artefacts [16]. Optical fibers integrated in FRP tendons are usually bonded by the resin, which is available intrinsically during the manufacturing of the tendons, and the adhesive thickness can be therefore neglected. However, under certain stress states, the measurement can be affected by the tendon material, as FRP have a low shear stiffness compared to metals. This leads to a pronounced non-uniform normal stress distribution under discontinuous shear stress boundary conditions [7]. In addition, these so-called shear lag effects are thought to be a limiting factor for the maximum tensile strength of FRP material when anchored using short mechanical anchorages with high shear stress gradients e.g. resin-grout anchors [21]. Depending on the shear stiffness of the FRP, the diameter of the rod and the radial position, the measured normal strain deviates from the value which would have been expected from a one-dimensional analysis, where a uniform stress over the whole cross-section is assumed.

Table 1
Properties of tendon material.

Property	Tendon
Tendon CFRP diameter d	7.5 mm $\pm$ 0.1 mm
Coated diameter	8.5 mm $\pm$ 0.3 mm
Sand coating	Quartz sand 0.1–0.6 mm
Carbon fiber	Teijin Tenax™ UTS50 F24 24K 1600tex [22]
Tensile strength fiber $\sigma_{f_{ult}}$	5100 MPa
Young's modulus fiber E_{f11}	245 GPa
Fiber volume fraction V_f	0.6
Resin system	Huntsman Aradur epoxy
Young's modulus resin E_m	2.85 GPa
Poisson's ratio ν_m	0.35
Optical fiber	Fibercore SM1250B3 (9.8/125) [23]

Table 2
Properties of concrete.

Property	Concrete
Cement CEM I 42.5 R	440 kg/m ³
Limestone powder	207 kg/m ³
Alluvial aggregates 0–1 mm	609 kg/m ³
Alluvial aggregates 1–4 mm	498 kg/m ³
Alluvial aggregates 4–8 mm	487 kg/m ³
Water	176 kg/m ³
Superplasticizer (polycarboxylate-based, dry content: 40 mass-%)	5.28 kg/m ³
w/c	0.4
w/b	0.27
Flow	660 mm
Fresh density	2360 kg/m ³
Air content	3.4%
Compressive strength at 28 d EN12390-3:2009; 150 × 150 × 150 mm ³	79.5 $\pm$ 2.95 MPa
Young's modulus at 28 d EN 12390-13:2021; 120 × 120 × 360 mm ³	38.3 $\pm$ 0.63 GPa

2. Methods

The main focus of this study was on experiments, for which pull-out tests of CFRP tendons with integrated DFOS were cast and tested. In parallel, a finite element model was developed, which allowed to assess the quality of the modeled pull-out behavior using τ - δ relationships. Another question which arises when analyzing strain data of integrated DFOS is shear lag. Due to the limited possibility to experimentally prove this effect, the main focus was here on the numerical modeling.

2.1. Tendons

In this study, sand-coated CFRP tendons with a nominal diameter d of 7.5 mm were studied. The tendon material properties are summarized in Table 1. The quartz sand coating with grain size 0.1–0.6 mm was adhesively bonded using the same resin system as for the production of the CFRP tendon. The polyimide coated optical fiber of type Fibercore SM1250B3(9.8/125) with a cladding diameter of 125 μ m and an outer diameter of 155 μ m was integrated and extracted according to [19]. The centrality of the optical fiber cannot be assumed to be perfect but with a maximum eccentricity of 2 mm. This was proven on representative samples using microscopic images.

2.2. Concrete

A self-compacting high performance concrete (HPC) of strength class C60/75 with binder composed of a blend of Portland cement CEM I 42.5R (68% by mass of binder) and limestone powder (32% by mass of binder) was used. The concrete contained aggregate with sizes up to 8 mm. The fresh concrete properties, the compressive strength and the elastic modulus at 28 days are also reported in Table 2.



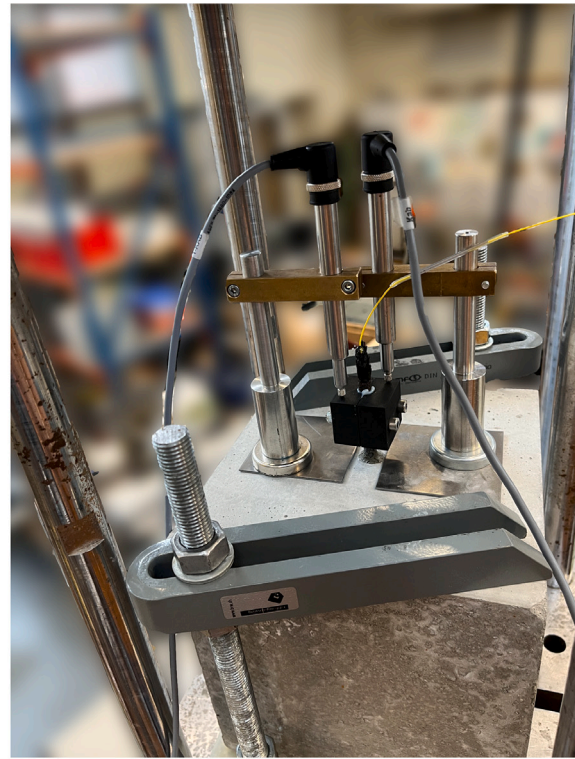
Fig. 1. Formwork with tendon and bond breaker, the bare optical fiber can be seen on the left hand side.

2.3. Pull-out tests

In total 9 pull-out specimens with different embedment length were cast. Casting direction and specimen geometry were selected according to standard DIN EN 10080:2005 [8]. The cube dimensions were 200 × 200 × 200 mm³ and the embedment length was varied between 5 · $d \approx$ 40 mm, 10 · $d \approx$ 80 mm and 15 · $d \approx$ 120 mm. For the bond breaker, a polymethylmethacrylate (PMMA) tube with an inner diameter of 10 mm and outer diameter of 15 mm was used (see Fig. 1). The sand-coated tendon was sealed against the PMMA tube using a kneading compound (Bostik GmbH, Borgholzhausen, Germany), providing a watertight seal with negligible shear stresses on the tendon during loading. After casting, the cubes were left in moulds for 7 days and stored in a climate conditioned room at >90% RH and 20 °C. The cubes were demoulded at the age of 7 days and stored at room temperature. The fragile optical fibers were spliced together with pigtails, which are necessary for connecting to the fiber optic interrogator. The brittle nature of the uncoated optical fiber made splicing delicate and partial failure was hardly avoidable. The pull-out tests were performed after 28 days. The machine used for testing was a servo-hydraulic universal test machine Instron 1343 with a capacity of 250 kN (see Fig. 2(a)). Specimens were loaded in a displacement-controlled manner with a speed of 1 mm/min by the stroke of the testing machine. On the actuator side, a mechanical wedge anchorage developed by [19] was used to transfer the actuator force to the tendon. On the unloaded end of the embedment section, the slippage measurement with respect to the concrete surface was not possible according to the standard, as the optical fiber was extracted at the end of the tendon (see Fig. 2(b)). To overcome this problem, two-part polymer clamps were adhesively bonded and additionally clamped with screws. The slippage was measured using two inductive half-bridge transducers of type Hottinger W5TK with a measuring range of $\pm$ 5 mm. The second and optional displacement transducer was used to reduce potential measurement errors resulting from the eccentric position of the axis of the transducers. The transducers were fixed on the concrete block using adhesively bonded steel plates and magnets, which allowed proper alignment of the transducers. Clamping claws fixating the concrete cube were necessary, as the cube could potentially lift off after a sudden slippage of the tendon. The strains in the optical fiber were measured using an optical distributed sensor interrogator ODISI 6104 from LUNA Inc. [24]. The gage pitch was selected to 0.65 mm and the measurement frequency was 32.5 Hz.



(a) Testing machine with the whole set-up installed.



(b) Unloaded side with displacement transducers.

Fig. 2. Experimental set-up.

2.4. Analytical model

The differential equation of bond can be derived based on an equilibrium of forces of the tendon and the surrounding concrete. If linear elastic material is assumed and by the definition of the slip $\delta = u_t - u_c$, where u_t is the displacement of the tendon and u_c is the displacement of the concrete, the following equation can be obtained:

$$\frac{d^2\delta}{dx^2} = \frac{d\epsilon_t}{dx} - \frac{\epsilon_c}{dx} = \frac{4 \cdot \tau}{d \cdot E_t} + \frac{d \cdot \pi \cdot \tau + q_x}{A \cdot E_c \cdot (1 - \rho_t)}, \quad (1)$$

where E_t is the Young's modulus of the tendon and E_c of the concrete, respectively. q_x is a potential line distributed force along the axis of the tendon.

By assuming a small reinforcement content ρ_t , which is justified by the huge concrete cross section compared to the small tendon diameter, the second term can be neglected and the following simpler equation remains:

$$\frac{d^2\delta}{dx^2} = \frac{d\epsilon_t}{dx} = \frac{4 \cdot \tau}{d \cdot E_t}. \quad (2)$$

Analytical solutions only exist for very simple functions of $\tau(x)$, which are often assumed to be a function of the slip $\tau(\delta(x))$. Nonetheless, (2) allows to calculate the local bond stress $\tau(x)$ by the following expression:

$$\tau(x) = \frac{d \cdot E_t}{4} \frac{d\epsilon_t(x)}{dx}. \quad (3)$$

2.5. Finite element model

One way to solve the differential equation of bond for general τ - δ -laws are numerical methods. Here, a finite element (FE) analysis was done using the commercial FE software Abaqus/CAE 2024. The complexity of the problem was reduced by assuming rotational symmetry with respect to the axis of the tendon. Due to the cubic

shape of the concrete block, this simplification does not reflect the reality. However, the modeling approach can be justified by the fact that the load transfer mechanism takes place in the narrow region of the tendon and that the additional concrete cover is only necessary to avoid concrete splitting. Therefore, the diameter of the concrete cylinder was modeled to be 225.7 mm resulting in a similar cross sectional area as for the concrete cube. The dimensions of both parts, the tendon and the concrete, are shown in Fig. 3. 4-node bilinear axisymmetric quadrilateral elements of type CAX4 with an element size of 0.5 mm were used for both domains, concrete and tendon. A mesh convergence study ranging from an element size of 2 mm to 0.25 mm was performed and showed that the chosen element size is small enough to guarantee a high solution quality. The tendon was modeled with anisotropic material properties according to Table 3. The concrete was assumed to show linear-elastic behavior with a Young's modulus after 28 days according to Table 2. The Poisson's ratio of the concrete was selected as 0.15, in accordance with the literature [25]. Boundary conditions were applied on the tendon's axis to avoid rigid body motion. In addition, the right side of the concrete cylinder was horizontally constrained. The load on the tendon was defined by a displacement in order to improve numerical stability and thus to avoid convergence problems. The τ - δ -behavior was modeled using a contact with cohesive properties, as proposed by [26]. [15] on the other hand, used cohesive elements instead of cohesive contacts, which was not followed here due to potential convergence problems resulting from highly distorting elements.

The linear elastic traction-separation behavior available in Abaqus can be written in matrix form:

$$\mathbf{t} = \begin{Bmatrix} t_n \\ t_s \\ t_t \end{Bmatrix} = \begin{bmatrix} K_{nn} & K_{ns} & K_{nt} \\ K_{ns} & K_{ss} & K_{st} \\ K_{nt} & K_{st} & K_{tt} \end{bmatrix} \begin{Bmatrix} \delta_n \\ \delta_s \\ \delta_t \end{Bmatrix} = \mathbf{K} \boldsymbol{\delta}, \quad (4)$$

where t_i denotes traction stress components, K_{ij} are the elasticity components and δ_i the separation [27]. The direction n corresponds

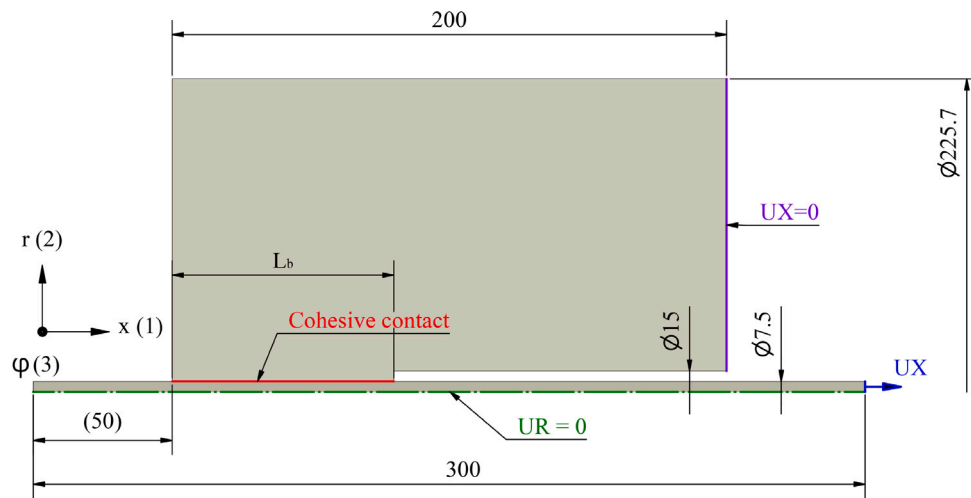


Fig. 3. Dimensions and boundary conditions of pull-out FE model.

to the normal direction (r -axis in Fig. 3) and s and t to the tangential direction (φ - and x -direction in Fig. 3).

Due to the axis-symmetry of the round tendon, coupling between tangential and radial as well as tangential and axial direction can be neglected. Therefore, uncoupled behavior can be studied, where the off-diagonal terms are zero and the formula simplifies to:

$$\begin{Bmatrix} t_n \\ t_s \\ t_t \end{Bmatrix} = \begin{Bmatrix} K_{nn} \cdot \delta_n \\ K_{ss} \cdot \delta_s \\ K_{tt} \cdot \delta_t \end{Bmatrix} \quad (5)$$

For the specific case of the τ - δ law, a unidirectional mechanism is usually assumed, where shear stresses are transferred in the longitudinal direction of the tendon, while normal stresses are evolving in normal direction, due to shrinkage and Poisson's effects. In order to model a stiff contact between the tendon and the sand coating, the elastic component K_{nn} was selected to be magnitudes higher (here $K_{nn} = 10^5$) than the elastic components in transverse direction K_{ss} and K_{tt} . The elastic shear stiffness in transverse direction was selected based on the τ - δ model. Even though only one of the two components is relevant for the bond stress transfer, both were set to the same value. Applying a zero or very low stiffness to the irrelevant direction could lead to numerical problems in 3D-models and should be avoided. In order to model softening of the bond, which inevitably happens during pull-out, an appropriate damage initiation criterion is necessary. The most suitable criterion for the modeling of τ - δ behavior in Abaqus is given by the maximum nominal stress criterion, where damage is initiated, when the following conditions is fulfilled:

$$\max \left\{ \frac{|t_n|}{t_n^0}, \frac{t_s}{t_s^0}, \frac{t_t}{t_t^0} \right\} = 1, \quad (6)$$

where t_i^0 denote the peak values of the nominal stresses, when loaded either in normal direction ($i = n$) or in shear direction ($i = s, t$). The nominal peak stresses in normal direction are not directly related to the mechanism of τ - δ damage. Therefore again, t_n^0 was selected magnitudes higher (here $t_n^0 = 10^5$) than the shear components t_s^0 and t_t^0 , which were selected based on the value of the initial stiffness K_i and the corresponding τ - δ model. Once the initiation criterion is fulfilled, softening of the contact can be modeled by a damage evolution law. This constitutive response is described by the τ - δ model. For that reason, a damage variable D is introduced, which models the degradation of the contact stiffness. For the damage variable D has to hold that $D \in [0, 1]$. The slip at initiation of damage is defined by $\delta_0 > 0$. δ_0 can be interpreted as the linear limit, up to where Abaqus assumes a linear traction–separation response. Furthermore, the additional condition $D(\delta_0) = 0$ must hold.

Damage evolution is handled in Abaqus by:

$$\begin{aligned} t_n &= \begin{cases} (1-D)\bar{t}_n & \text{if } \bar{t}_n \geq 0 \\ \bar{t}_n & \text{otherwise} \end{cases} \\ t_s &= (1-D)\bar{t}_s \\ t_l &= (1-D)\bar{t}_l, \end{aligned} \quad (7)$$

where $\tilde{t}_i$ are the stresses predicted by the elastic traction–separation behavior for the current strains without damage [27]. Clearly, negative normal traction stresses do not lead to damage and are directly transferred. If only the relevant shear component is studied, this can be rewritten for arbitrary τ - δ laws:

$$\tau(\delta) = (1 - D(\delta)) \cdot K_i \cdot \delta, \quad \delta \geq \delta_0. \quad (8)$$

K_i is the initial stiffness of the τ - δ curve and corresponds to the relevant elasticity component of (5). By rearranging (8), the damage variable D can be determined by the following expression:

$$D(\delta) = \begin{cases} 0 & \text{if } \delta \in [0, \delta_0] \\ 1 - \frac{\tau(\delta)}{K \cdot \delta} & \text{if } \delta \geq \delta_0 \end{cases} \quad (9)$$

The value of the initial stiffness K_i is constant and has to fulfill the following condition:

$$K_i \geq \frac{\tau(\delta)}{\delta}, \quad \delta \geq \delta_0 \quad (10)$$

This means in other words that the initial stiffness curve defined by K_i crosses the bond-slip only once at the location δ_0 . Multiple tangential contacts with the τ - δ curve would also be accepted. But if it would cross the τ - δ curve more than once, negative D would result, which cannot be handled by the FE solver. Generally, the stiffness K_i should be maximized while still fulfilling condition (10) in order to reduce deviations from a measured or modeled τ - δ curve, especially for small slips δ .

2.6. Shear lag of FRP

Shear lag is an integral part of the strain distribution measured during the pull-out test. A dedicated FE model was set up for an isolated study of those effects.

2.6.1. Materials

In addition to CFRP, the shear lag behavior was also studied for steel and GFRP, which are most commonly found in structural engineering as reinforcement materials. Steel was assumed to have linear elastic properties defined by the Young's modulus $E = 210$ GPa and the Poisson's ratio of $\nu = 0.3$. For CFRP and GFRP on the other hand, anisotropic

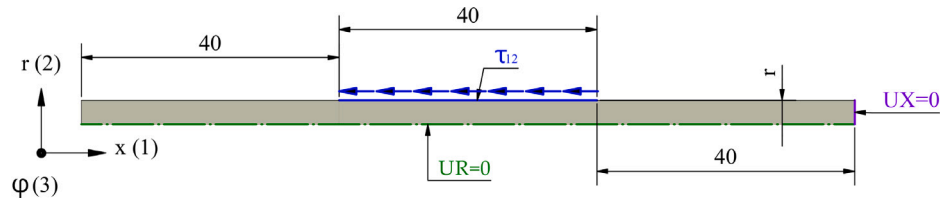


Fig. 4. Geometry and boundary conditions of shear lag FE model.

Table 3

(a) Properties of constituent fibers and matrix [28]. (b) Calculated composite constants according to [28].

(a) Fiber and matrix.			
Property	Unit	–	–
<i>Fiber</i>	–	<i>Carbon</i>	<i>Glass</i>
E_{f11}	GPa	245 [22]	73.1
$E_{f22} = E_{f33}$	GPa	13.79	73.1
$G_{f12} = G_{f13}$	GPa	8.96	30.1
G_{f23}	GPa	4.83	30.1
$\nu_{f12} = \nu_{f13}$	–	0.20	0.22
ν_{f23}	–	0.25	0.22
<i>Matrix</i>	–	<i>Epoxy</i>	
E_m	GPa	2.85	
ν_m	–	0.35	
(b) Composite.			
Property	Unit	–	–
<i>Composite</i>	–	<i>CFRP</i>	<i>GFRP</i>
V_f	–	60%	
E_{11}	GPa	148.1	45.0
$E_{22} = E_{33}$	GPa	7.39	11.15
$G_{12} = G_{13}$	GPa	3.33	4.18
G_{23}	GPa	2.67	4.18
$\nu_{12} = \nu_{13}$	–	0.260	0.272
ν_{23}	–	0.383	0.334

material constants reported by Chamis were used [28] (see Table 3). The fiber volume fraction V_f was assumed to be 60%, which is a typical value obtained in tape-laying processes and reflecting the tendons used in this study. In pultrusion processes, fiber volume fractions up to 70% can be reached [29]. Higher fiber volume fractions will reduce shear lag effects. The assumption of $V_f = 60\%$ is therefore rather conservative. The fiber, matrix and composite properties are shown in Table 3. For CFRP, E_{f11} was selected 245 GPa for Teijin Tenax™ UTS50, while the other material properties were assumed following [28], where Torayca T300 fibers were considered.

2.6.2. FE modeling

Similar to the model for the pull-out tests, an axisymmetric model was used. In accordance, CAX4 elements were used with a variation of the element size, as shown subsequently. The geometry and the boundary conditions were applied as illustrated in Fig. 4. The tendon was loaded by a uniform shear stress τ_{21} . The dimensions of the three segments were selected to be as short as possible to save computational time but still being long enough to avoid significant boundary edge effects. This was validated by the uniformity of the strain distribution close to the boundaries. The radius r was varied, such that the influence of different tendon diameters could be studied.

3. Results

3.1. Pull-out tests

In the following, bond-slip relationships are presented with a focus on average bond-stress vs. end-slip curves, the strain distribution in the ascending branch of the bond-slip curve, local bond-stress vs. slip

curves and the strain distribution in the descending branch of the bond-slip curve.

3.1.1. τ_{avg} vs. δ_{end} -curves

The quantitative evaluation of conventional pull-out experiments is usually limited to τ_{avg} vs. end-slip δ_{end} curves, whereas τ_{avg} is calculated as:

$$\tau_{avg} = \frac{F_{11}}{d \cdot \pi \cdot L_b} = \frac{\sigma_{11} \cdot d}{4 \cdot L_b}, \quad (11)$$

with the pull-out force F_{11} , the embedment length L_b and the normal stress σ_{11} resulting from F_{11} .

Fig. 5(a) shows in total 9 curves, three per embedment length studied. The measurement range of 10 mm was limited by the displacement transducer. All specimens with embedment length 40 mm showed a continuous softening. In contrast to them, all specimens with embedment length 120 mm and one with embedment length 80 mm showed sudden pull-out failure without significant softening. Due to the elastic energy stored in the tendon, large displacements were observed, for some specimen even with a complete release of the force. Due to the high energy release during the jerk, the mounted polymer clamps, acting as reference for the displacement transducer, broke, for which reason the measurement was interrupted. Fig. 5(b) focusses on the region of small δ_{end} , where in case of sand-coated FRP τ_{avg} is maximal. Longer embedment lengths tend to show smaller τ_{avg} , which is well-known [7,12]. Shorter embedment lengths in particular show lower initial stiffness, which can be explained by the strain distribution, highlighted subsequently. One representative specimen per embedment length is studied in detail below. The selected specimens are plotted in Figs. 5(a) and 5(b) in thick continuous lines.

The typical failure patterns of the tendons are shown in Fig. 6(a), mainly delamination or often called peeling of the sand coating was observed also by [30,31]. However for all tendons, cement paste fractures were found on the loaded side of the embedment length, close to the bond breaker, which indicates a partial pull-out failure. Here, the shear strength of the adhesively bonded sand-coating overpassed the strength of the concrete. This delamination was also reflected in the strain distributions measured, as shown below. By visual analysis of the embedment length in the concrete cubes in Figs. 6(b)–6(d), a significant amount of delaminated CFRP fibers was found. This means that the tendon delaminated between the CFRP rather than simply peeling off the sand-coating. Moreover, the pattern of the tendons and the embedment sections in the concrete correlate well. In addition, it can be proven that no cement paste penetrated the bond breaker, which underlines the effectiveness of sealing.

3.1.2. Ascending branch of τ_{avg} vs. δ_{end} curve

The standard approach for studying the evolution of the bond in the embedment length is given by the strain distribution for different normal stresses of the tendon. This is done in the following for one specimen per embedment length. Another interesting plot arises, when the strain distribution $\epsilon_{11}(x)$ is normalized by the normal stress at the loaded end:

$$C_{11}(x) = \frac{\epsilon_{11}(x)}{\sigma_{11}}. \quad (12)$$

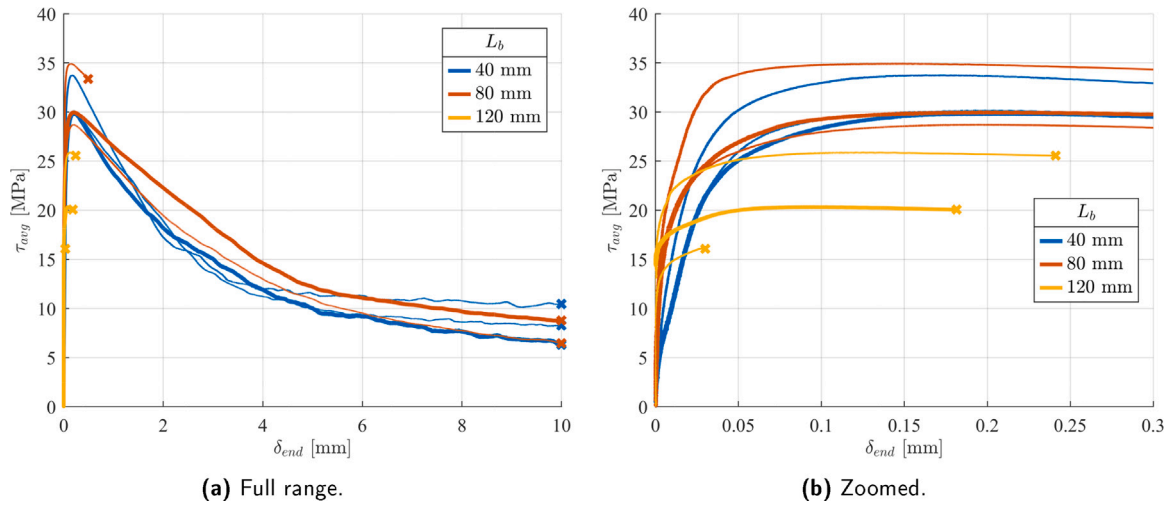
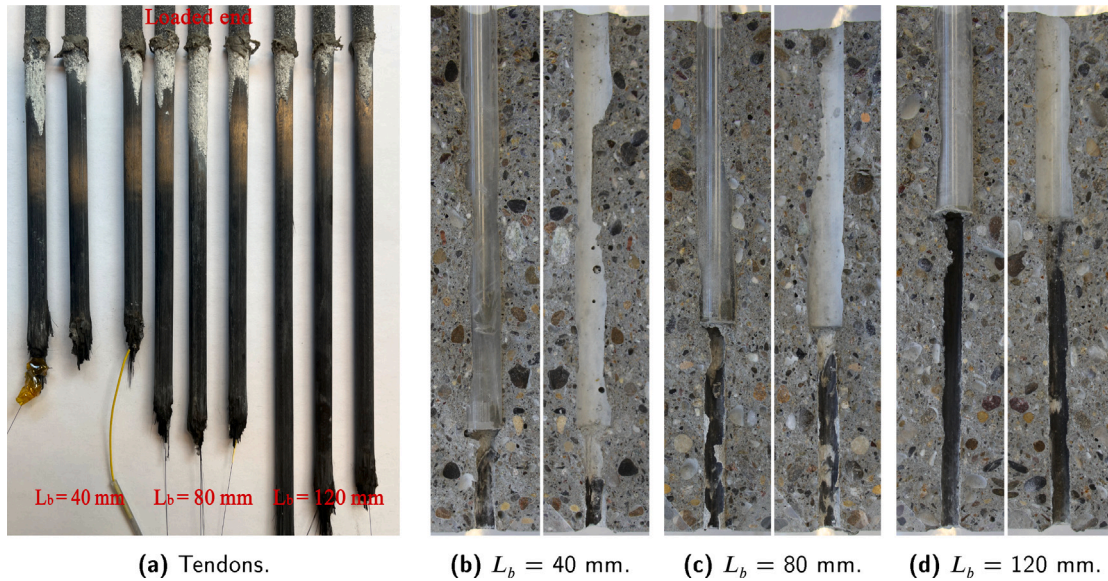
Fig. 5. τ_{avg} - δ_{end} curves.

Fig. 6. Visual analysis with partially delaminated CFRP tendon and partial pull-out failure of the concrete.

The values in the free length of the tendon correspond then to the inverse of the Young's modulus, and thus $C_{11} = \frac{1}{E_{11}}$. Those illustrations in particular allow to follow how the bond is activated and how the bond stress evolves from the loaded to the unloaded end. Fig. 7(a) shows the strain distribution for an embedment length of 40 mm. The maximal normal stress was reached at $\sigma_{11} = 637$ MPa. The plateau at the loaded end of the embedment section is thought to result from the pulled-out section with cement paste (see Fig. 6(a)).

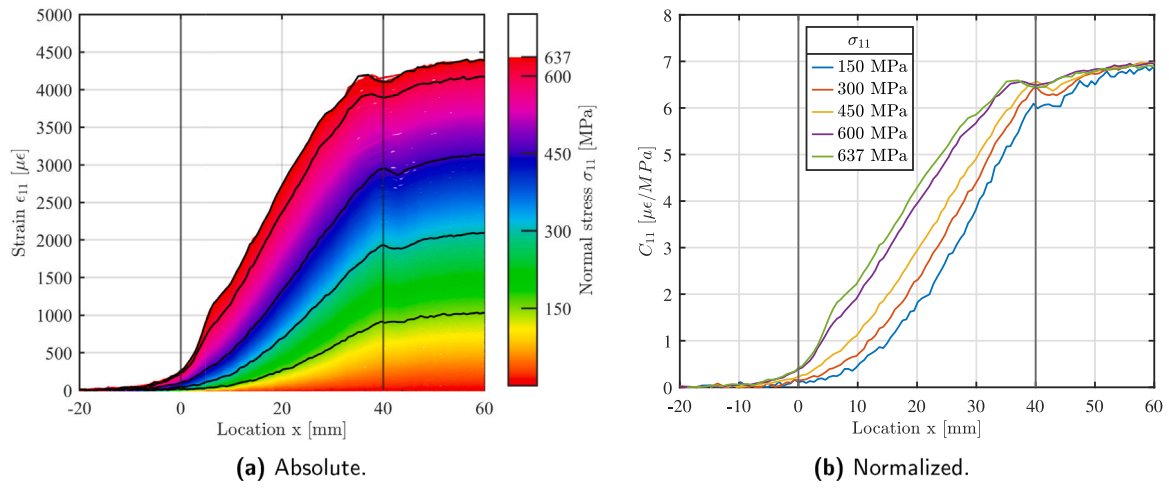
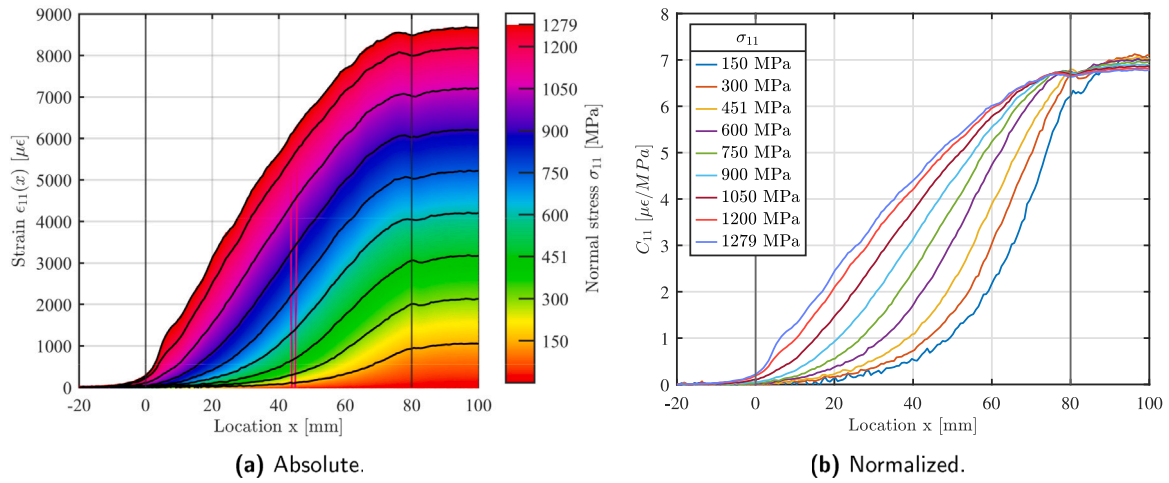
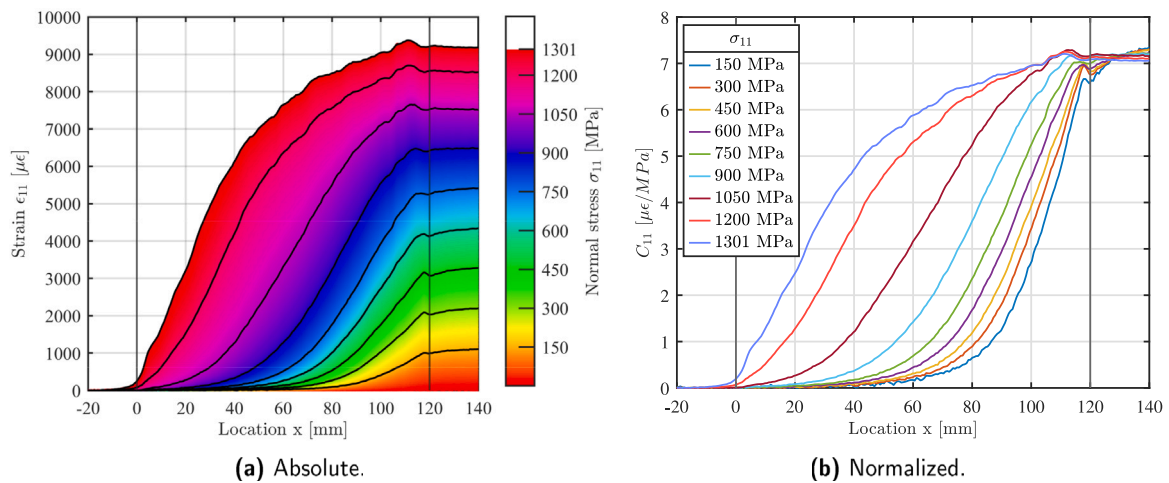
Fig. 7(b) reveals that the strain is close to linear in the mid of the embedment length, especially at normal stresses close to the maximal normal stress. However, the strain is already close to constant at the loaded end of the embedment length, reflecting local pull-out failure (see Fig. 6(a)). On the other hand, for smaller σ_{11} , the embedment length is not fully activated, meaning that the region close to the unloaded end is not substantially contributing to the shear stress transfer and the strain distribution shows a positive curvature. The normal strain at the unloaded end tends to be slightly positive. This can be explained by shear lag, as shown in Section 3.2. This effect is more pronounced for high shear stress discontinuities, which increase during the activation of bond regions located close to the unloaded end due to the increasing load. This behavior was observed independently from the

embedment lengths, as shown subsequently. Fig. 8(a) shows the strain distribution for the specimen with an embedment length of 80 mm. The maximum normal stress was 1279 MPa and therefore roughly double with respect to the embedment length of 40 mm.

The normalized strain distribution in Fig. 8(b) shows a similar curve shape as for the embedment length of 40 mm. Initially the curve has a positive curvature and tends then to a negative curvature, especially at points located close to the loaded end of the embedment length. At higher normal stresses, a linear strain distribution in the mid section can be interpreted. The horizontal plateau at the loaded end is less pronounced as in Fig. 7(b), which correlates well with the damage pattern in Fig. 6(a).

Finally, Fig. 9(a) shows the strain distribution for an embedment length of 120 mm. Apparently, the maximum normal stress is insignificantly higher than for the 80 mm embedment length, even though the embedment length is 50% longer. This can be explained by the pronounced concavity of the strain distribution, which is due to delamination of the CFRP tendon (see Fig. 6(d)).

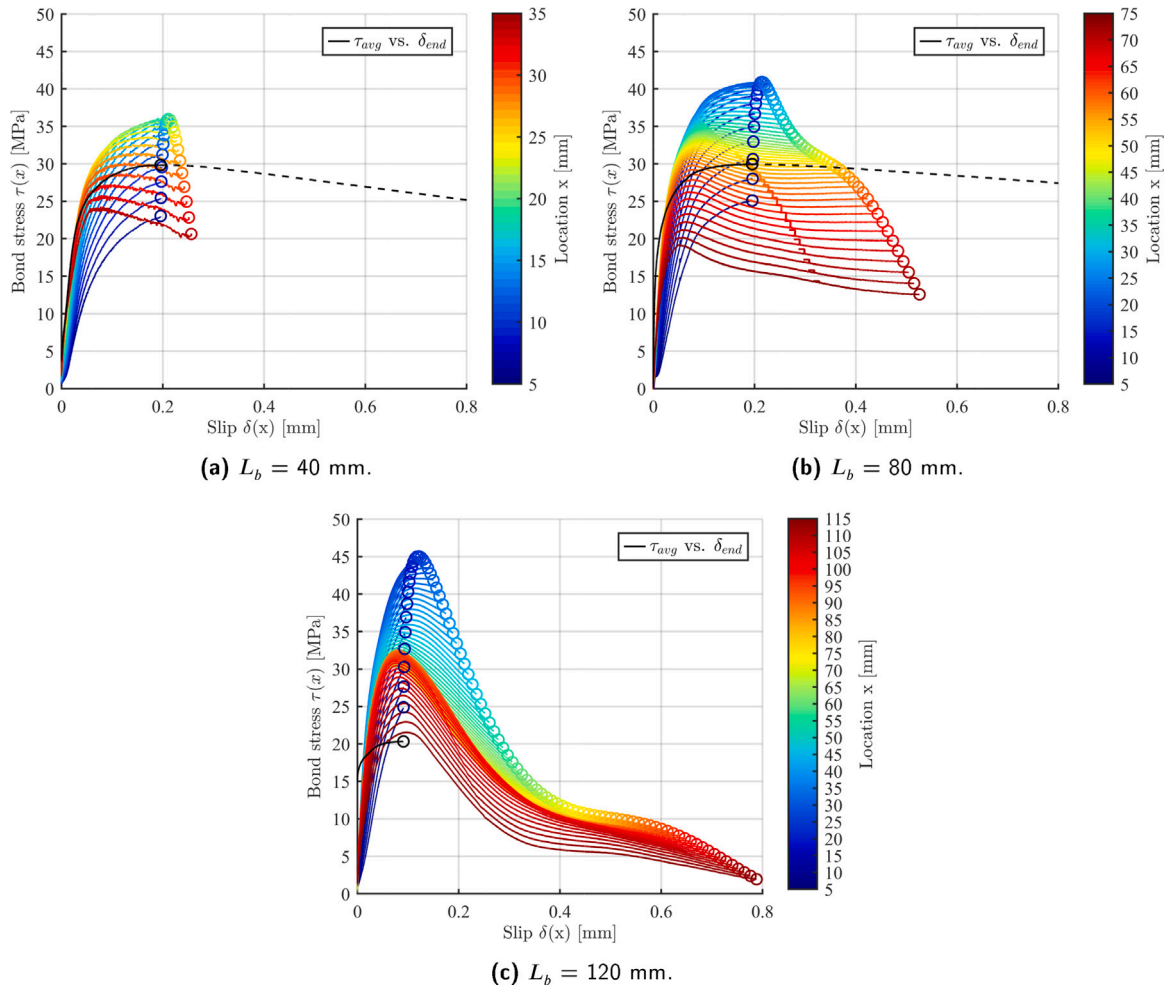
The normalized strain distribution in Fig. 9(b) shows again the evolution and activation of the bond. Initially, only the right half of the embedment length is activated by a strain distribution with pronounced

Fig. 7. Strain distribution for $L_b = 40$ mm.Fig. 8. Strain distribution for $L_b = 80$ mm.Fig. 9. Strain distribution for $L_b = 120$ mm.

positive curvature. Later, at the maximum normal stress, the situation is vice versa. The right side of the embedment length shows a negative curvature, equivalent to a decreasing bond stress. In contrast, the left half of the embedment length shows an almost linear distribution.

3.1.3. Local τ - δ curves

Local τ - δ curves can be obtained if strain distribution curves are differentiated and integrated. Details about the post-processing are given in [Appendix](#). [Figs. 10\(a\)–10\(c\)](#) show local τ - δ curves. Points closer than

Fig. 10. Local τ vs. δ .

5 mm to the boundaries of the embedment length were excluded, as small errors in the polynomial approximation of the strain distribution could lead to significant deviations in the bond stress. For comparison, the black line, which is partially dashed, shows the τ_{avg} - δ_{end} curve measured by the load cell and the displacement transducer. The circles describe the points at maximum average bond strength. By the calculation of the mean of the local bond stresses, τ_{avg} would be obtained in good agreement. This is not surprising, but confirms that the assumptions made are correct. For short embedment lengths (see Fig. 10(a)), the end slip still correlates well with the local slips. In addition, local τ reach the highest values of approximately 35 MPa in the center of the embedment length. For embedment lengths of 80 mm (Fig. 10(b)), the differences in the local slip are already significant and are not well-approximated by δ_{end} . Moreover, the maximum bond stresses are reached in the back third. Values of up to 40 MPa were derived based on the slope of the strain distribution. For the embedment length of 120 mm (Fig. 10(c)), again the highest bond stresses are reached at the back third of the embedment length, with values of approximately 45 MPa. Local slips reach almost 0.8 mm, while the measured δ_{end} at maximum τ_{avg} reaches only 0.1 mm. The radical decrease of the bond stress close to the loaded end is obvious. This again can be well explained by the partial pull-out failure, where the concrete is sheared off.

3.1.4. Descending branch of τ_{avg} vs. δ_{end} curve

After reaching the maximum average bond strength, the τ - δ behavior is usually characterized by a linear descending branch and followed by a horizontal line [32]. Due to the significant end slip, coordinates of

the tendon need to be adjusted, as the focus still lies on the bond stress in the embedment length. Therefore, the coordinates x are translated by δ_{end} :

$$\tilde{x} = x - \delta_{end} \quad (13)$$

Figs. 11(a) and 11(b) show the strain distribution for an embedment length of 40 mm and 80 mm, respectively. Both measurements show a very steep increase at the unloaded end of the section. This could be due to the intact sand-coated part of the tendon, which is getting pulled in. The effect of the protruding bar portions at the unloaded end on the pull-out force was also reported by [33]. On the other end of the embedment length, the strain distribution is almost horizontal, equivalent to non-existing bond stresses. Since concrete is partially sheared off in this section, the real embedment length is shorter than initially. For both specimens studied, the 20 mm of the embedment length located next to the loaded end show missing bond stresses after end slip δ_{end} larger than 5 mm. Apart from the aforementioned details, the strain distribution can be roughly approximated by a linear function. In this section, the sand-coating is peeled off and small inter-laminar shear stresses can therefore be assumed. The slope of the strain distribution and thus the bond stress decreases continuously with increasing δ_{end} .

3.1.5. FE-Solution

Figs. 12(a) and 12(b) show the τ_{avg} - δ_{end} curves obtained for embedment lengths of 40 and 80 mm. The black lines show the experimentally measured values, identical to Fig. 5. The blue circles show the input

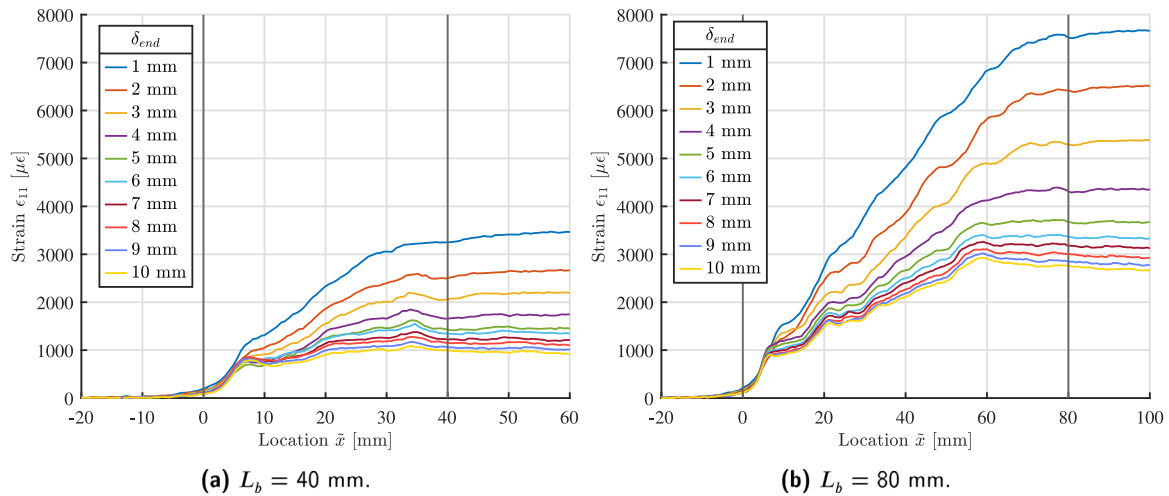
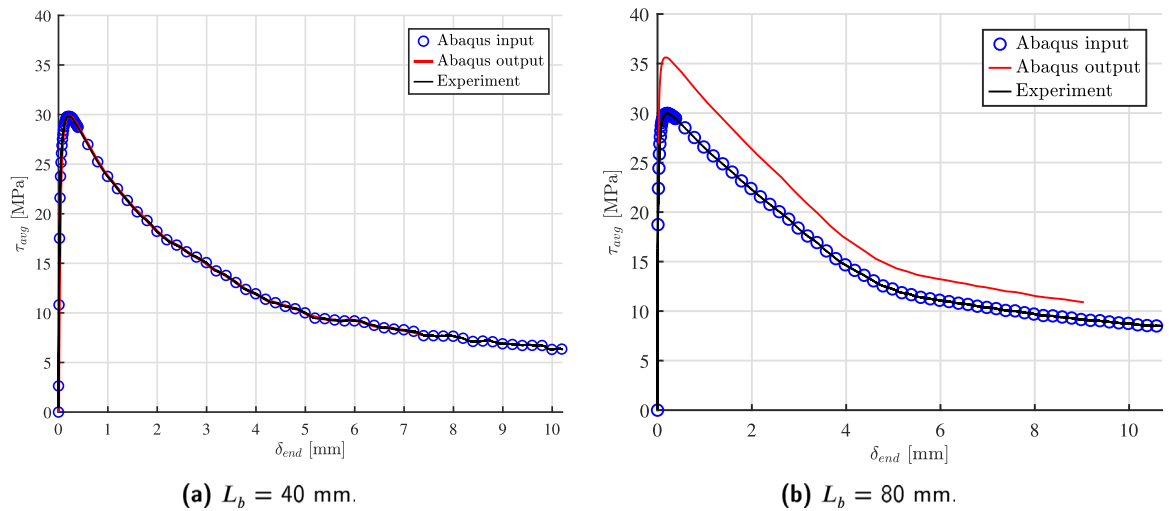


Fig. 11. Strain distribution of descending branch.

Fig. 12. τ_{avg} - δ_{end} curve FE vs. experiment.

data for the FE calculation taking into account the reformulation of the τ_{avg} - δ_{end} law based on the damage model, as described in Section 2.5. Obviously, the bond-slip law can be accurately approximated also with an initial linear stiffness and subsequent damage evolution for both embedment lengths. For $L_b = 40$ mm, the output of the FE analysis (red line) almost perfectly matches with the input curve. This is in accordance with Fig. 10(a), where a small variation in the slip δ was reported. For $L_b = 80$ mm on the other hand, the FE analysis overestimated the bond strength consistently over the whole slip range.

When focussing on the strain distribution, larger deviations are reported, as potentially expected from Fig. 12(a). Fig. 13(a) reveals that the τ_{avg} - δ_{end} law overestimates the bond stress near the loaded end and underestimates the bond stress near the unloaded end of the embedment length. On average, those two effects cancel each other out. Again, the assumed τ_{avg} - δ_{end} law is not able to model the local pull-out failure at the loaded end of the embedment length, which is thought to be the main source of the deviations. This is also a limitation of the FE model, which assumes a linear elastic response of the concrete. As a result, damage is not taken into account.

For the embedment length of 80 mm, the deviations are even more pronounced (see Fig. 13(b)). Obviously, the approximation of the measured strain distribution is not accurate. In accordance with Fig. 13(a), only curves with identical normal stresses σ_{11} were plotted. As it can be concluded from Fig. 12(b), the maximum normal stress σ_{11}

reached 1520 MPa in the FE analysis instead of the 1279 MPa measured in the experiment.

3.2. Shear Lag

The axial stress distribution at the central axis of the tendon for a shear stress τ_{21} is shown in Fig. 14. Instead of normal strains, normal stresses were selected due to the different Young's modulus E_{11} of the materials studied. The black dashed line corresponds to a one-dimensional model, with a single degree of freedom per node and therefore purely axial stiffness. This curve corresponds also to the widely used model of constant bond stress. Steel, with a relatively high shear stiffness, shows only a negligible shear lag. Also GFRP with an intermediate shear stiffness shows moderate stress variations. On the other hand, CFRP with its very low shear stiffness shows significant shear lag, which spreads up to ± 20 mm from the point of discontinuous boundary conditions at $x = 0$ and $x = 40$ mm. Shear lag is always related to a boundary condition defined by a shear stress. Those shear stresses are in equilibrium with axial stresses (see Fig. 14). To solely analyze the effects of the shear lag stresses, those normal stresses have to be subtracted. Therefore an isolated stress resulting from the shear

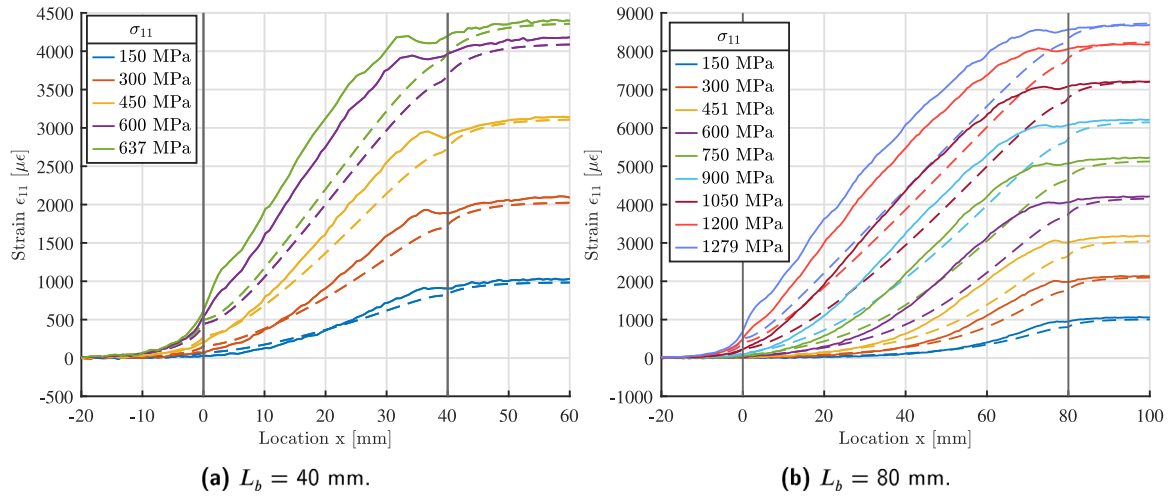
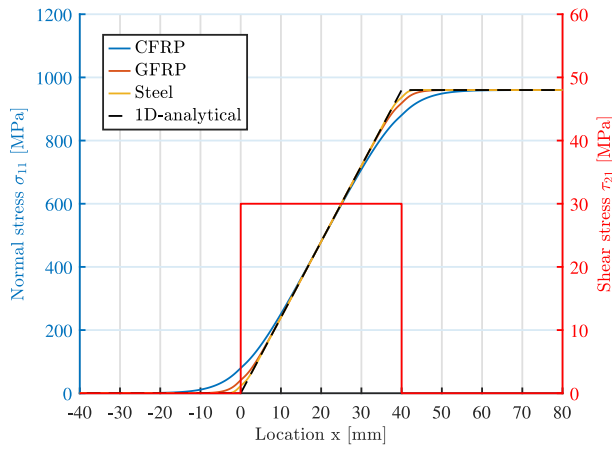
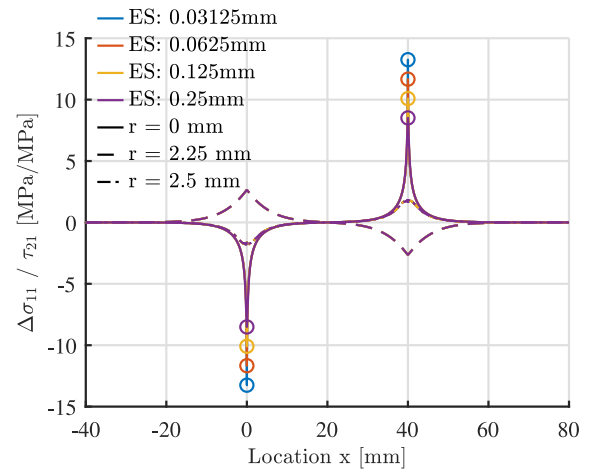


Fig. 13. Strain distribution: FE (dashed) vs. experiment.

Fig. 14. $\sigma_{11}(r = 0)$ vs. location for tendon $d = 5$ mm.Fig. 15. Mesh convergence for CFRP with $d = 5$ mm.

lag effect can be defined by:

$$\Delta\sigma_{11} = \begin{cases} \sigma_{11} & \text{if } x < 0 \\ \sigma_{11} - \frac{4 \cdot \tau_{21} \cdot x}{d} & \text{if } x \in [0, L_b] \\ \sigma_{11} - \frac{4 \cdot \tau_{21} \cdot L_b}{d} & \text{if } x > L_b. \end{cases} \quad (14)$$

For the integral of $\Delta\sigma_{11}$ over the cross-sectional area holds that the equilibrium is fulfilled for every location x :

$$F_{11} = \iint_A \Delta\sigma_{11}(r, \varphi) dr d\varphi = 0. \quad (15)$$

For the quantification of the shear lag, a consistent embedment length L_b of 40 mm was selected. Due to the linear assumption of the underlying finite element model, a linear relationship is also given between σ_{11} , $\Delta\sigma_{11}$ and the shear stress τ_{21} . Fig. 15 shows normalized shear lag normal stress distributions for different radial positions. Apparently, local symmetries with respect to the edges of the boundary conditions at $x = 0$ mm, and $x = 40$ mm can be verified. Furthermore, a point symmetry in the center of the definition of the shear stress boundary conditions at $x = 20$ mm is observed. Normal stresses related to the shear lag in the center of the tendon are positive at the unloaded end and negative at the loaded end. For points located on the surface the opposite applies, in accordance with (15).

3.2.1. Stress singularity

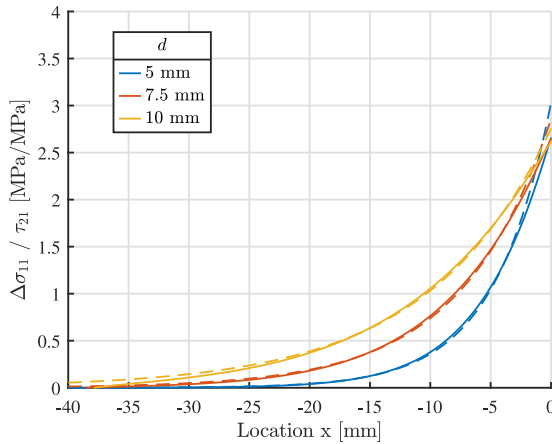
Sharp corners, point forces, point restraints and other discontinuities are all sources of so-called stress singularities. Evidently, also shear lag resulting from discontinuous boundary conditions can result in such a singularity phenomenon. Fig. 15 shows the stress variations for element sizes of 0.25, 0.125, 0.0625 and 0.03125 mm. Element sizes were intentionally selected as a multiple of the smallest element size. Strains are plotted for locations at the center ($r = 0$ mm), close to the outer surface ($r = 2.25$ mm) and at the surface ($r = 2.5$ mm), where the boundary condition was applied. Mesh dependence is exclusively apparent for elements located at the surface and only at the point of discontinuous shear stress, as confirmed by Table 4. Further reducing the mesh size leads to increasing stresses, which confirms the presence of a stress singularity. Points located at $r = 2.25$ mm and 0 mm show no mesh dependence and can also be accurately approximated by practically relevant mesh sizes of 0.25 mm. The slightly deviating value at $r = 2.25$ mm with an element size of 0.25 mm is related to the fact that the studied node is part of the outermost element, which is affected by the stress singularity. For the further studies, an element size of 0.125 mm was selected.

3.2.2. Axial strain distribution

For practical reasons, an analytical approximation of the shear lag would be beneficial. This would allow to roughly quantify its effects

Table 4Mesh convergence for CFRP with $d = 5$ mm at $x = 0$ mm.

Mesh size	$\Delta\sigma_{11}/\tau_{21}$ [MPa/MPa]		
	$r = 0$ mm	$r = 2.25$ mm	$r = 2.5$ mm
0.03125 mm	2.6539	1.7618	-13.2531
0.0625 mm	2.6539	1.7620	-11.6677
0.125 mm	2.6539	1.7612	-10.0856
0.25 mm	2.6539	1.8380	-8.5114

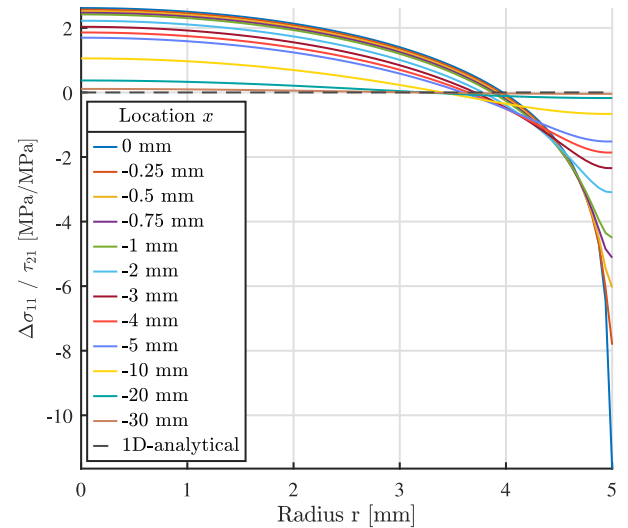
**Fig. 16.** $\Delta\sigma_{11}/\tau_{21}$ vs. location for CFRP, dashed lines for LLS, solid lines for FE.**Table 5**Shear lag approximation model for $r = 0$ mm.

	CFRP		
	$d = 5$ mm	$d = 7.5$ mm	$d = 10$ mm
$\frac{\sigma_{11}}{\tau_{21}}$ (FE)	2.6539	2.6458	2.6149
m (LLS)	3.0273	2.8557	2.7527
n (LLS)	4.70 mm	7.41 mm	10.20 mm
	GFRP		
	$d = 5$ mm	$d = 7.5$ mm	$d = 10$ mm
$\frac{\sigma_{11}}{\tau_{21}}$ (FE)	1.3803	1.3803	1.3802
m (LLS)	1.6090	1.6072	1.6069
n (LLS)	2.10 mm	3.21 mm	4.28 mm

and act as a simple tool for the analysis of results. By logarithmic plotting of $\Delta\sigma_{11}/\tau_{21}$ vs. the location x , it is found that the curve can be approximated by a function of the shape:

$$\frac{\Delta\sigma_{11}}{\tau_{21}} = m \cdot \exp\left(\frac{x}{n}\right), \quad (16)$$

where m and n are constants. The constants can be determined based on a linear least square (LLS) approximation. Here, locations x between -20 and 0 mm were selected for the definition of the constants m and n . Fig. 16 shows the approximation (dashed lines) vs. the strain distributions obtained by the FE analysis (solid lines). The LLS overestimates the shear lag effect for $x > -3$ mm and also for locations $x < -25$ mm. In-between a good correlation can be found. Table 5 summarizes the values obtained for the constants for both, CFRP and GFRP and the three different tendon diameters studied. By comparing the numerical values, it can be proven that the constant m overestimates the shear lag at $x = 0$. Interestingly, the shear lag at $x = 0$ is almost independent of the tendon diameter. However, the decay rate characterized by the constant n highly depends on the diameter of the tendon, as expected. The shear lag stress is approximately double for CFRP compared to GFRP, and the decay rate behaves similarly. The decay rate is roughly a linear function of the diameter of the tendon.

**Fig. 17.** $\Delta\sigma_{11}/\tau_{21}$ vs. radius r for CFRP with $d = 10$ mm.

3.2.3. Radial strain distribution

Due to the equilibrium of forces (15), which has to be fulfilled for every cross-sectional area, an inhomogeneous radial strain field results. Fig. 17 shows the radial strain distribution for different locations x for a CFRP tendon with a diameter of 10 mm. Clearly, the stress singularity is rediscovered at $x = 0$ mm. In addition, the decay of the inhomogeneous radial distribution for locations away from the initiation point of the stress singularity is obvious. At a distance of 4 mm, they have already halved and at distances of 20 mm, they have almost completely subsided. Assuming that the optical fiber is not perfectly located at the center of the tendon, smaller shear lag effects can be taken into account. If the optical fiber would be located close to the surface, even the sign would change.

4. Conclusions

The present study considers strain distributions of sand-coated CFRP tendons during direct pull-out tests, which are widely used for the characterization of the bond between reinforcement and concrete. The strain distribution was measured using integrated distributed fiber optic sensing. This allowed to calculate local bond stresses and slip, while measuring also the average bond stress and the end slippage. This allows to draw the following conclusions:

- Integrated DFOS have been proven to be mature for measuring high quality strain distributions in pull-out tests with minimal need for data filtering. Furthermore, the bond is not affected by the sensing technique. Nonetheless, the laborious preparation of the optical fiber has to be pointed out.
- The longer the embedment length, the higher are the local bond stresses. This is in contradiction with the average bond stress, which is smaller, the longer the embedment length is. Maximum local bond stresses are located in the center or closer to the unloaded end of the embedment section.
- Local bond-slip curves are position dependent. This explains why local strain distributions cannot be well-approximated by average bond-stress vs. end-slip curves, especially for long embedment lengths. On the other hand, the formal analysis with position dependent bond-slip curves is not reasonable, due to the complexity of the physical mechanism involved. Additionally, no apparent correlation is found between the curves obtained from different embedment lengths.

- The shorter the embedment length, the better the approximation of the local bond–slip behavior by average bond-stress vs. end-slip curves. This can be explained by the smaller deviation of the local slips from the end slippages.
- Depending on the concrete strength, partial pull-out failure can affect the embedment length. Average bond-stresses obtained from pull-out experiments cannot directly be applied for other embedment lengths or loading conditions. Local bond–slip models or more sophisticated concrete material models taking into account damaged plasticity could capture this effect.
- Strain distributions calculated by finite element methods based on average bond-stress vs. end-slip curves show significant differences to the measured curves. This underlines that average bond-stress vs. end-slip curves cannot directly be applied for precise bond modeling.
- It was experimentally and numerically proven that fiber reinforced polymers show significant shear lag effects. The lower the shear stiffness is, the larger the shear lag effects. In the same way, the larger the tendon diameter, the more pronounced the shear lag effects. The axial strain distribution at the centerline of the tendon can be approximated by an exponential function. The maximal stress deviation is material-dependent but almost independent of the radius of the tendon. However, the decay rate depends mainly on the diameter of the tendon. Those shear lag effects have to be taken into account when analyzing measurements of integrated optical fibers.
- The stress distribution due to shear lag is a function of the radial position, rod diameter and material. Exact experimental quantification of the shear lag in experiments is only possible if the exact position of the optical fiber is known. Depending on the radial position, the stresses resulting from shear lag can be either positive or negative. In addition, the stress concentration at the radius of the perturbation has been identified as stress singularity. This stress singularity leads to a non-uniform stress state and will reduce the theoretical tensile strength of FRP. Smaller diameter, higher fiber volume contents and smaller bond stresses can mitigate those shear lag effects.

The knowledge gained from the local bond–slip states opens new research questions. One question is: how can measured bond–slip curves be applied for the prediction of transfer lengths when considering pre-tensioned applications? Another question is: how has a bond–slip law to be defined, that is able to depict the true strain distribution in pull-out experiments and how does this correlate then to the local bond–slip curves, which are found to be location-dependent? In general, measurements using integrated DFOS allow to further understand the bond evolution and the stability of bond between FRP and concrete.

CRediT authorship contribution statement

Valentin Ott: Writing – original draft, Software, Investigation, Formal analysis, Conceptualization. **Mateusz Wyrzykowski:** Writing – review & editing, Resources, Investigation. **Pietro Lura:** Writing – review & editing, Supervision, Resources. **Christian Affolter:** Writing – review & editing, Software. **Giovanni Pietro Terrasi:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Post-processing of strain distribution

The post-processing of strain distribution, especially the numerical differentiation of strain distributions bring a high risk of misinterpretation of results. Therefore, further informations are given on how the curves were post-processed and what limitations could arise. The specific case of an embedment length of 80 mm was selected for illustration.

A.1. Strain distribution

The first derivative of the strain distribution is required for the calculation of the bond stress. Even though integrated optical fibers allow to measure much smoother strain distributions than externally bonded optical fibers, they are still not smooth enough for the calculation of a derivative. Therefore, a polynomial fitting using MATLAB R2023b was done. The curve was fitted in a range of $x = [-20, L_b + 20]$ mm. Polynomials with 8th order showed a satisfying approximation of the strain distribution (see Fig. 18(a)). Another advantage of polynomial functions are their continuity and the existence of exact derivatives, which simplifies the calculation of bond stresses.

A.2. Local slip calculation

The local slip $\delta(x)$ is the sum of the end slip δ_{end} and the integral of the strains in the range between 0 and x :

$$\delta(x) = \delta_{end} + \int_0^x \epsilon_{11}(\eta) d\eta. \quad (17)$$

As long as no strain spikes are part of the domain of interest, exact integration can simply be replaced by a numerical integration through a sum of the strains. Noisy strain measurements require always filtering, as shown by [16]. The slip distribution for different normal stress levels is shown in Fig. 18(b). The contribution of the end slip δ_{end} can be clearly seen at the location $x = 0$ mm, especially for higher σ_{11} . For $\sigma_{11} = 900$ MPa, the local slip δ_x is already larger than 0.2 mm while the end slip is still close to 0.

A.3. Local bond stress calculation

The local bond stress can be calculated according to (3). As already stated above, the higher curvature of the strain distribution at the boundaries of the embedment length can affect the quality of the polynomial fitting. Therefore, the range of confidence was limited to $x = [5, L_b - 5]$ mm. The local bond stresses τ for different normal stresses are shown in Fig. 18(c). Apparently, the bond stress is hardly constant in the whole range. The bond stress reaches higher values at the center than at the boundaries and the point of maximal bond stress is shifted to the unloaded end of the embedment length.

Data availability

Data will be made available on request.

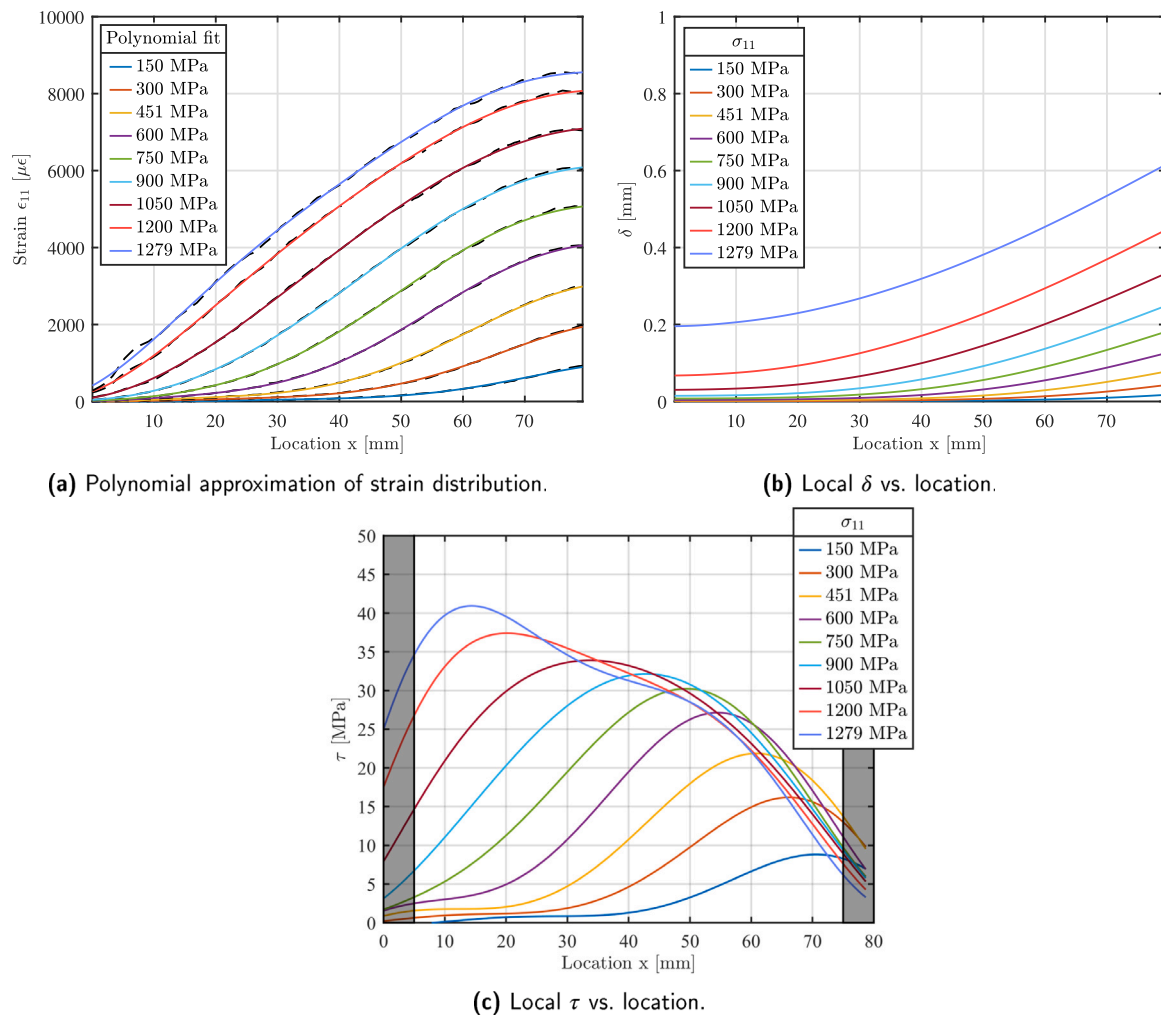


Fig. 18. Post-processing of strain distribution.

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